

LAB3.md

Script: enhanced_numbers.sh

Original Behavior

- `print_numbers.sh` printed numbers from 1 to 5 with a fixed step of 1.

New Behavior

- `enhanced_numbers.sh` allows the user to specify the start, end, and step values as command-line arguments.
- The script checks that the step is positive and prints numbers from start to end using the given step.

Example Runs

Valid Input:

```
$ ./enhanced_numbers.sh 2 10 2
2
4
6
8
10
```

```
vaibhavgkumar@VAIBHAVs-MacBook-Air ~ % nano backup.sh
vaibhavgkumar@VAIBHAVs-MacBook-Air ~ % chmod 777 backup.sh
vaibhavgkumar@VAIBHAVs-MacBook-Air ~ % ./backup.sh 2 10 2
2
4
6
8
10
vaibhavgkumar@VAIBHAVs-MacBook-Air ~ % █
```

```
1
1
1
```

Invalid Step:

```
$ ./enhanced_numbers.sh 1 5 0
Error: Step must be a positive integer.
```

Missing Arguments:

```
$ ./enhanced_numbers.sh 1 5
Usage: ./enhanced_numbers.sh <start> <end> <step>
```

Extra Questions

1. Difference between `$1`, `$@`, and `$#` in bash?

- `$1`: The first command-line argument.
- `$@`: All command-line arguments as separate words.
- `$#`: The number of command-line arguments.

2. What does `exit 1` mean in a script?

- It ends the script with a status code of 1, indicating an error.